

ICE 3107: Digital Signal Processing

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Lecture Outcomes

Understand the concept, properties, and applications of the one-sided Z-transform.

One-Sided Z-Transform

- The unilateral Z-transform of x(n):

$$X^{+}(z) = \sum_{n=0}^{\infty} x(n) z^{-n}$$

- Only values from: $n = 0$ *to* ∞ are considered.

One-Sided Z-Transform (Cont.)

Used for:

i. Causal systems
ii. Initial value problems
iii. Difference equations
iv. Discrete-time control systems

Example 3.5.1

Determine the one-sided z-transform of the signals:

- $x_1(n)=\{1,2,5,7,0,1\}$

Ans: $X_1^+(z) = 1 + 2z^{-1} + 5z^{-2} + 7z^{-3} + z^{-5}$

- $x_2(n)=\{1, 2, 5\uparrow, 7, 0, 1\}$

Ans: $X_2^+(z) = 5 + 7z^{-1} + z^{-3}$

Time Shift Property: x(n−k)

General Formula:

$$Z^{+}\{x(n-k)\} = z^{-k}X^{+}(z) + \sum_{m=1}^{k} x(-m)z^{-k+m}$$

In case x(n) is causal, then

$$Z^{+}\{x(n-k)\} = z^{-k}X^{+}(z)$$

Practice Problem 01

- Given the discrete-time signal:

$$x(n) = \left(\frac{1}{2}\right)^n u(n)$$

Use the one-sided Z-transform time-shifting property to determine:

$$Z^+\{x(n-2)\}$$

Ans:

$$Z^+\{x(n-2)\} = z^{-2}\frac{1}{1-\frac{1}{2}z^{-1}}$$

ROC: $|z| > \frac{1}{2}$

Example 3.5.2

Determine the one-sided z-transform of the signals:

(a) $x(n) = a^n u(n)$

(b) $x_1(n) = x(n-2)$ where $x(n) = a^n$

Ans:

(a) $$X^+(z) = \frac{1}{1 - az^{-1}}$$

(b) $$X_1^+(z) = \frac{z^{-2}}{1 - az^{-1}} + a^{-1}z^{-1} + a^{-2}$$

Time Shift Property: x(n+k)

General Formula:

$$Z^{+}\{x(n+k)\} = z^{k}X^{+}(z) - \sum_{m=0}^{k-1} x(m)z^{k-m}$$

Expanded Form

$$Z^{+}\{x(n+k)\} = z^{k}X^{+}(z) - z^{k}x(0) - z^{k-1}x(1)$$

Practice Problem 02

- Given the discrete-time signal:

$$x(n) = \left(\frac{1}{2}\right)^n u(n)$$

Use the one-sided Z-transform time-shifting property to determine:

$$Z^{+}\{x(n+1)\}$$

Ans: $\frac{1}{2} \cdot \frac{1}{1 - \frac{1}{2}z^{-1}}$

ROC: $|z| > \frac{1}{2}$

Practice Problem 03

- Use the one-sided Z-transform to solve the following difference equation with the given initial condition:

$$y(n+1)-0.5y(n)=x(n),\ n\geq 0$$

Given: $y(0)=2$

Determine the output transform Y(z) in terms of the input transform X(z).

Ans:

$$Y(z)=\frac{X(z)+2z}{z-0.5}$$

Practice Problem 04

A digital system receives the signal:

$$x(n)=(0.6)^n u(n)$$

Find:

i. $Z^+\{x(n-1)\}$

ii. $Z^+\{x(n+1)\}$

Ans:

i. $$Z^+\{x(n-1)\}=\frac{z^{-1}}{1-0.6z^{-1}}$$

i. $$Z^+\{x(n+1)\}=\frac{0.6}{1-0.6z^{-1}}$$

Practice Problem 05

Solve using one-sided Z-transform:

$$y(n)-0.5y(n-1)=u(n)$$

Given:

$$y(-1)=1$$

Ans: $y(n)=2u(n)-0.5(0.5)^n u(n)$

Thank You


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